



llama-4-scout-17b-16e-instruct

Text Generation • Meta

@cf/meta/llama-4-scout-17b-16e-instruct

Meta's Llama 4 Scout is a 17 billion parameter model with 16 experts that is natively multimodal. These models leverage a mixture-of-experts architecture to offer industry-leading performance in text and image understanding.

Model Info

Context	131,000 tokens
---------	----------------

Window	↗
--------	-------------------

Model Info

Terms and License	link ↗
Function calling ↗	Yes
Batch	Yes
Unit Pricing	\$0.27 per M input tokens, \$0.85 per M output tokens

Playground

Try out this model with Workers AI LLM Playground. It does not require any setup or authentication and an instant way to preview and test a model directly in the browser.

Launch the LLM Playground

Usage

✓ Worker - Streaming

TypeScript

```
export interface Env {  
  AI: Ai;  
}  
  
export default {  
  async fetch(request, env):  
    Promise<Response> {  
  
    const messages = [  
      { role: "system", content: "You are a  
friendly assistant" },  
      {  
        role: "user",  
        content: "What is the origin of the  
phrase Hello, World",  
      },  
    ];  
  
    const stream = await  
env.AI.run("@cf/meta/llama-4-scout-17b-16e-  
instruct", {  
      messages,  
      stream: true,  
    });
```

```
return new Response(stream, {  
  headers: { "content-type":  
"text/event-stream" },  
  });  
},  
} satisfies ExportedHandler<Env>;
```

✓ Worker

TypeScript

```
export interface Env {  
  AI: Ai;  
}  
  
export default {  
  async fetch(request, env):  
    Promise<Response> {  
  
    const messages = [  
      { role: "system", content: "You are a  
friendly assistant" },  
      {  
        role: "user",  
        content: "What is the origin of the  
phrase Hello, World",  
      },  
    ];  
  
    const response = await  
env.AI.run("@cf/meta/llama-4-scout-17b-16e-  
instruct", { messages });  
  
    return Response.json(response);  
  },  
} satisfies ExportedHandler<Env>;
```


▼ Python

Python

```
import os
import requests

ACCOUNT_ID = "your-account-id"
AUTH_TOKEN = os.environ.get("CLOUDFLARE_AUTH_TOKEN")

prompt = "Tell me all about PEP-8"
response = requests.post(
    f"https://api.cloudflare.com/client/v4/accounts/{ACCOUNT_ID}/llama-4-scout-17b-16e-instruct",
    headers={"Authorization": f"Bearer {AUTH_TOKEN}"},
    json={
        "messages": [
            {"role": "system", "content": "You are a helpful assistant."},
            {"role": "user", "content": prompt}
        ]
    }
)
result = response.json()
print(result)
```

▼ curl

```
curl
```

```
https://api.cloudflare.com/client/v4/accounts/llama-4-scout-17b-16e-instruct \
-X POST \
-H "Authorization: Bearer $CLOUDFLARE_AUTH_TOKEN" \
-d '{ "messages": [{ "role": "system", "content": "You are a helpful assistant." }, { "role": "user", "content": "Why is pizza so good" } ] }'
```

① OpenAI compatible endpoints

Workers AI also supports OpenAI compatible API endpoints for `/v1/chat/completions` and `/v1/embeddings`. For more details, refer to [Configurations](#).

Parameters

* indicates a required field

Input

• **object**

- **prompt** **string** required min 1

The input text prompt for the model to generate a response.

- **guided_json** **object**

JSON schema that should be fulfilled for the response.

- **response_format** **object**

- **type** **string**

- **json_schema**

- **raw** **boolean**

If true, a chat template is not applied and you must adhere to the specific model's expected formatting.

- **stream** **boolean**

If true, the response will be streamed back incrementally using SSE, Server Sent Events.

- `max_tokens` **integer** default 256

The maximum number of tokens to generate in the response.

- `temperature` **number** default 0.15 min 0 max 5

Controls the randomness of the output; higher values produce more random results.

- `top_p` **number** min 0 max 2

Adjusts the creativity of the AI's responses by controlling how many possible words it considers. Lower values make outputs more predictable; higher values allow for more varied and creative responses.

- `top_k` **integer** min 1 max 50

Limits the AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises.

- **seed** **integer** min 1 max 9999999999

Random seed for reproducibility of the generation.

- **repetition_penalty** **number** min 0 max 2

Penalty for repeated tokens; higher values discourage repetition.

- **frequency_penalty** **number** min 0 max 2

Decreases the likelihood of the model repeating the same lines verbatim.

- **presence_penalty** **number** min 0 max 2

Increases the likelihood of the model introducing new topics.

- ▶ **1** **object**

- **messages** **array** required

An array of message objects representing the conversation history.

- **items** **object**

- **role** **string**

The role of the message sender (e.g., 'user', 'assistant', 'system', 'tool').

- `tool_call_id` **string**

The tool call id. If you don't know what to put here you can fall back to 000000001

- `content` **one of**

- `0` **string**

The content of the message as a string.

- `1` **array**

- `items` **object**

- `type` **string**

Type of the content provided

- `text` **string**

- `image_url` **object**

- `url` **string**

image uri with data (e.g.
...).
HTTP URL will not be accepted

- 2 **object**

- type **string**

Type of the content provided

- text **string**

- image_url **object**

- url **string**

image uri with data (e.g.
...).
HTTP URL will not be accepted

- functions **array**

- items **object**

- name **string** required

- code **string** required

- **tools** **array**

A list of tools available for the assistant to use.

- **items** **one of**

- **0** **object**

- **name** **string** required

The name of the tool. More descriptive the better.

- **description** **string** required

A brief description of what the tool does.

- **parameters** **object** required

Schema defining the parameters accepted by the tool.

- **type** **string** required

The type of the parameters object (usually 'object').

- **required** **array**

List of required parameter names.

- **items** **string**

- **properties** **object** required

Definitions of each parameter.

- **additionalProperties** **object**

- **type** **string** required

The data type of the parameter.

- **description** **string** required

A description of the expected parameter.

- **1** **object**

- **type** **string** required

Specifies the type of tool (e.g., 'function').

- **function** **object** required

Details of the function tool.

- **name** **string** required

The name of the function.

- **description** **string** required
A brief description of what the function does.
- **parameters** **object** required
Schema defining the parameters accepted by the function.
 - **type** **string** required
The type of the parameters object (usually 'object').
- **required** **array**
List of required parameter names.
 - **items** **string**
- **properties** **object** required
Definitions of each parameter.
 - **additionalProperties** **object**
 - **type** **string** required

The data type of the parameter.

- **description** **string** required

A description of the expected parameter.

- **response_format** **object**

- **type** **string**

- **json_schema**

- **guided_json** **object**

JSON schema that should be fulfilled for the response.

- **raw** **boolean**

If true, a chat template is not applied and you must adhere to the specific model's expected formatting.

- **stream** **boolean**

If true, the response will be streamed back incrementally using SSE, Server Sent Events.

- `max_tokens` **integer** default 256

The maximum number of tokens to generate in the response.

- `temperature` **number** default 0.15 min 0 max 5

Controls the randomness of the output; higher values produce more random results.

- `top_p` **number** min 0 max 2

Adjusts the creativity of the AI's responses by controlling how many possible words it considers. Lower values make outputs more predictable; higher values allow for more varied and creative responses.

- `top_k` **integer** min 1 max 50

Limits the AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises.

- `seed` **integer** min 1 max 9999999999

Random seed for reproducibility of the generation.

- `repetition_penalty` **number** min 0 max 2

Penalty for repeated tokens; higher values discourage repetition.

- `frequency_penalty` **number** min 0 max 2

Decreases the likelihood of the model repeating the same lines verbatim.

- `presence_penalty` **number** min 0 max 2

Increases the likelihood of the model introducing new topics.

- 2 **object**

- `requests` **array** required

- `items` **one of**

- 0 **object**

- `prompt` **string** required min 1

The input text prompt for the model to generate a response.

- `guided_json` **object**

JSON schema that should be fulfilled for the response.

- **response_format** **object**

- **type** **string**

- **json_schema**

- **raw** **boolean**

If true, a chat template is not applied and you must adhere to the specific model's expected formatting.

- **stream** **boolean**

If true, the response will be streamed back incrementally using SSE, Server Sent Events.

- **max_tokens** **integer** default 256

The maximum number of tokens to generate in the response.

- **temperature** **number** default 0.15 min 0 max 5

Controls the randomness of the output; higher values produce more random results.

- `top_p` **number** min 0 max 2

Adjusts the creativity of the AI's responses by controlling how many possible words it considers. Lower values make outputs more predictable; higher values allow for more varied and creative responses.

- `top_k` **integer** min 1 max 50

Limits the AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises.

- `seed` **integer** min 1 max 9999999999

Random seed for reproducibility of the generation.

- `repetition_penalty` **number** min 0 max 2
Penalty for repeated tokens; higher values discourage repetition.
- `frequency_penalty` **number** min 0 max 2
Decreases the likelihood of the model repeating the same lines verbatim.
- `presence_penalty` **number** min 0 max 2
Increases the likelihood of the model introducing new topics.
- 1 **object**
 - `messages` **array** required
An array of message objects representing the conversation history.
 - `items` **object**
 - `role` **string**
The role of the message sender (e.g., 'user', 'assistant', 'system', 'tool').

- `tool_call_id` **string**

The tool call id. If you don't know what to put here you can fall back to 000000001

- `content` **one of**

- `0` **string**

The content of the message as a string.

- `1` **array**

- `items` **object**

- `type` **string**

Type of the content provided

- `text` **string**

- `image_url` **object**

- `url` **string**

image uri with data
(e.g.
...). HTTP URL
will not be accepted

- 2 **object**

- type **string**

Type of the content
provided

- text **string**

- image_url **object**

- url **string**

image uri with data (e.g.
data:image/jpeg;base64,
/9j/...). HTTP URL will not
be accepted

- functions **array**

- items **object**

- **name** **string** required

- **code** **string** required

- **tools** **array**

A list of tools available for the assistant to use.

- **items** **one of**

- **0** **object**

- **name** **string** required

The name of the tool. More descriptive the better.

- **description** **string** required

A brief description of what the tool does.

- **parameters** **object** required

Schema defining the parameters accepted by the tool.

- **type** **string** required

The type of the parameters object (usually 'object').

- **required** **array**

List of required parameter names.

- **items** **string**

- **properties** **object** required

Definitions of each parameter.

- **additionalProperties**

object

- **type** **string** required

The data type of the parameter.

- **description** **string**

required

A description of the expected parameter.

■ 1 **object**

■ type **string** required

Specifies the type of tool (e.g., 'function').

■ function **object** required

Details of the function tool.

■ name **string** required

The name of the function.

■ description **string** required

A brief description of what the function does.

■ parameters **object** required

Schema defining the parameters accepted by the function.

- **type** **string** required

The type of the parameters object (usually 'object').

- **required** **array**

List of required parameter names.

- **items** **string**

- **properties** **object**

required

Definitions of each parameter.

- **additionalProperties** **object**

- **type** **string**

required

The data type of the parameter.

- **description**
string required
A description of the expected parameter.
- **response_format** **object**
 - **type** **string**
 - **json_schema**
- **guided_json** **object**
JSON schema that should be fulfilled for the response.
- **raw** **boolean**
If true, a chat template is not applied and you must adhere to the specific model's expected formatting.
- **stream** **boolean**
If true, the response will be streamed back incrementally using SSE, Server Sent

Events.

- `max_tokens` **integer** default 256

The maximum number of tokens to generate in the response.

- `temperature` **number** default 0.15 min 0 max 5

Controls the randomness of the output; higher values produce more random results.

- `top_p` **number** min 0 max 2

Adjusts the creativity of the AI's responses by controlling how many possible words it considers. Lower values make outputs more predictable; higher values allow for more varied and creative responses.

- `top_k` **integer** min 1 max 50

Limits the AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values

introduce more variety and potential surprises.

- **seed** **integer** min 1 max 9999999999

Random seed for reproducibility of the generation.

- **repetition_penalty** **number** min 0 max 2

Penalty for repeated tokens; higher values discourage repetition.

- **frequency_penalty** **number** min 0 max 2

Decreases the likelihood of the model repeating the same lines verbatim.

- **presence_penalty** **number** min 0 max 2

Increases the likelihood of the model introducing new topics.

Output

- **0** **object**

- **response** **string** required

The generated text response from the model

- **usage** **object**

Usage statistics for the inference request

- **prompt_tokens** **number** 0

Total number of tokens in input

- **completion_tokens** **number** 0

Total number of tokens in output

- **total_tokens** **number** 0

Total number of input and output tokens

- **tool_calls** **array**

An array of tool calls requests made during the response generation

- **items** **object**

- **id** **string**

The tool call id.

- **type** **string**

Specifies the type of tool (e.g., 'function').

- **function** **object**

Details of the function tool.

- **name** **string**

The name of the tool to be called

- **arguments** **object**

The arguments passed to be passed to the tool call request

- ▶ **1** **string**

API Schemas

The following schemas are based on JSON Schema

Input

Output

```
{
  "type": "object",
  "oneOf": [
    {
      "title":
"Ai_Cf_Meta_Llama_4_Prompt",
      "properties": {
        "prompt": {
          "type": "string",
          "minLength": 1,
          "description": "The input
text prompt for the model to generate a
response."
        },
        "guided_json": {
          "type": "object",
          "description": "JSON
schema that should be fulfilled for the
response."
        },
        "response_format": {
          "title": "JSON Mode",
          "type": "object",
          "properties": {
```

```
        "type": {
          "type": "string",
          "enum": [
            "json_object",
            "json_schema"
          ]
        },
        "json_schema": {}
      },
      "raw": {
        "type": "boolean",
        "default": false,
        "description": "If true, a
chat template is not applied and you must
adhere to the specific model's expected
formatting."
      },
      "stream": {
        "type": "boolean",
        "default": false,
        "description": "If true,
the response will be streamed back
incrementally using SSE, Server Sent Events."
      },
```

```
"max_tokens": {  
  "type": "integer",  
  "default": 256,  
  "description": "The  
maximum number of tokens to generate in the  
response."  
},  
"temperature": {  
  "type": "number",  
  "default": 0.15,  
  "minimum": 0,  
  "maximum": 5,  
  "description": "Controls  
the randomness of the output; higher values  
produce more random results."  
},  
"top_p": {  
  "type": "number",  
  "minimum": 0,  
  "maximum": 2,  
  "description": "Adjusts  
the creativity of the AI's responses by  
controlling how many possible words it  
considers. Lower values make outputs more
```


predictable; higher values allow for more varied and creative responses."

```
    },  
    "top_k": {  
      "type": "integer",  
      "minimum": 1,  
      "maximum": 50,  
      "description": "Limits the
```

AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises."

```
    },  
    "seed": {  
      "type": "integer",  
      "minimum": 1,  
      "maximum": 9999999999,  
      "description": "Random  
seed for reproducibility of the generation."  
    },
```

```
    "repetition_penalty": {  
      "type": "number",  
      "minimum": 0,  
      "maximum": 2,
```

```
        "description": "Penalty  
for repeated tokens; higher values discourage  
repetition."  
    },  
    "frequency_penalty": {  
        "type": "number",  
        "minimum": 0,  
        "maximum": 2,  
        "description": "Decreases  
the likelihood of the model repeating the same  
lines verbatim."  
    },  
    "presence_penalty": {  
        "type": "number",  
        "minimum": 0,  
        "maximum": 2,  
        "description": "Increases  
the likelihood of the model introducing new  
topics."  
    }  
},  
"required": [  
    "prompt"  
]  
},
```

```
{
  "title":
"Ai_Cf_Meta_Llama_4_Messages",
  "properties": {
    "messages": {
      "type": "array",
      "description": "An array
of message objects representing the
conversation history.",
      "items": {
        "type": "object",
        "properties": {
          "role": {
            "type":
"string",
            "description":
"The role of the message sender (e.g., 'user',
'assistant', 'system', 'tool')."
          },
          "tool_call_id": {
            "type":
"string",
            "description":
"The tool call id. If you don't know what to
put here you can fall back to 000000001",
```

```
        "pattern": "[a-zA-Z0-9]{9}"
      },
      "content": {
        "oneOf": [
          {
            "type": "string",

            "description": "The content of the message as
a string."
          },
          {
            "type": "array",

            "items": {
              "type": "object",

              "properties": {
                "type": {
                  "type": "string",
```

```
"description": "Type of the content provided"

},

"text": {

"type": "string"

},

"image_url": {

"type": "object",

"properties": {

"url": {

"type": "string",

"pattern": "^data:*",

"description": "image uri with data (e.g.
```

...). HTTP URL will not be accepted"

}

}

}

}

}

},

{

"type": "object",

"properties": {

"type": {

"type": "string",

"description": "Type of the content provided"

},

"text": {

```
"type": "string",  
  
    },  
  
    "image_url": {  
  
        "type": "object",  
  
        "properties": {  
  
            "url": {  
  
                "type": "string",  
  
                "pattern": "^data:*",  
  
                "description": "image uri with data (e.g.  
...). HTTP URL will  
not be accepted"  
  
            }  
  
        }  
  
    }  
  
}
```

```
    }  
  ]  
}  
}  
}  
},  
"functions": {  
  "type": "array",  
  "items": {  
    "type": "object",  
    "properties": {  
      "name": {  
        "type":  
"string"  
      },  
      "code": {  
        "type":  
"string"  
      }  
    },  
    "required": [  
      "name",  
      "code"  
    ]  
  }  
},  
"required": [  
  "name",  
  "code"  
]  
}
```



```
    },  
    "tools": {  
      "type": "array",  
      "description": "A list of  
tools available for the assistant to use.",  
      "items": {  
        "type": "object",  
        "oneOf": [  
          {  
            "properties":  
  
            {  
              "name": {  
  
                "type": "string",  
  
                "description": "The name of the tool. More  
descriptive the better."  
              },  
  
              "description": {  
  
                "type": "string",  
  
                "description": "A brief description of what  
the tool does."
```

```
    },  
  
    "parameters": {  
  
      "type": "object",  
  
      "description": "Schema defining the parameters  
accepted by the tool.",  
  
      "properties": {  
  
        "type": {  
  
          "type": "string",  
  
          "description": "The type of the parameters  
object (usually 'object')."  
        },  
  
        "required": {  
  
          "type": "array",  
  
          "description": "List of required parameter  
names.",
```

```
"items": {  
  
  "type": "string"  
}  
  
},  
  
"properties": {  
  
  "type": "object",  
  
  "description": "Definitions of each  
parameter.",  
  
  "additionalProperties": {  
  
    "type": "object",  
  
    "properties": {  
  
      "type": {  
  
        "type": "string",
```

```
"description": "The data type of the  
parameter."
```

```
},
```

```
"description": {
```

```
"type": "string",
```

```
"description": "A description of the expected  
parameter."
```

```
}
```

```
},
```

```
"required": [
```

```
"type",
```

```
"description"
```

```
]
```

```
}  
  
    },  
  
    "required": [  
  
    "type",  
  
    "properties"  
    ],  
  },  
  "required": [  
    "name",  
  
    "description",  
  
    "parameters"  
  ],  
},  
{  
  "properties":  
  
  "type": {
```

```
"type": "string",

"description": "Specifies the type of tool
(e.g., 'function')."

    },

"function": {

"type": "object",

"description": "Details of the function
tool.",

"properties": {

"name": {

"type": "string",

"description": "The name of the function."
    },

"description": {
```

```
"type": "string",

"description": "A brief description of what
the function does."

},

"parameters": {

"type": "object",

"description": "Schema defining the parameters
accepted by the function.",

"properties": {

"type": {

"type": "string",

"description": "The type of the parameters
object (usually 'object')."

},


```

```
"required": {  
  
  "type": "array",  
  
  "description": "List of required parameter  
names.",  
  
  "items": {  
  
    "type": "string"  
  
  }  
  
},  
  
"properties": {  
  
  "type": "object",  
  
  "description": "Definitions of each  
parameter.",  
  
  "additionalProperties": {
```



```
"type": "object",

"properties": {

  "type": {

    "type": "string",

    "description": "The data type of the
parameter."

  },

  "description": {

    "type": "string",

    "description": "A description of the expected
parameter."

  }

},
```

```
"required": [
```

```
"type",
```

```
"description"
```

```
]
```

```
}
```

```
}
```

```
},
```

```
"required": [
```

```
"type",
```

```
"properties"
```

```
]
```

```
}
```

```
},
```

```
"required": [  
  
  "name",  
  
  "description",  
  
  "parameters"  
  
],  
}  
},  
"required": [  
  "type",  
  "function"  
],  
}  
]  
}  
},  
"response_format": {  
  "title": "JSON Mode",  
  "type": "object",  
  "properties": {  
    "type": {  
      "type": "string",
```

```
        "enum": [  
            "json_object",  
            "json_schema"  
        ],  
    },  
    "json_schema": {}  
},  
"guided_json": {  
    "type": "object",  
    "description": "JSON  
schema that should be fulfilled for the  
response."  
},  
"raw": {  
    "type": "boolean",  
    "default": false,  
    "description": "If true, a  
chat template is not applied and you must  
adhere to the specific model's expected  
formatting."  
},  
"stream": {  
    "type": "boolean",  
    "default": false,
```

```
      "description": "If true,
the response will be streamed back
incrementally using SSE, Server Sent Events."
    },
    "max_tokens": {
      "type": "integer",
      "default": 256,
      "description": "The
maximum number of tokens to generate in the
response."
    },
    "temperature": {
      "type": "number",
      "default": 0.15,
      "minimum": 0,
      "maximum": 5,
      "description": "Controls
the randomness of the output; higher values
produce more random results."
    },
    "top_p": {
      "type": "number",
      "minimum": 0,
      "maximum": 2,
```

`"description": "Adjusts the creativity of the AI's responses by controlling how many possible words it considers. Lower values make outputs more predictable; higher values allow for more varied and creative responses."`

`},`

`"top_k": {`

`"type": "integer",`

`"minimum": 1,`

`"maximum": 50,`

`"description": "Limits the AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises."`

`},`

`"seed": {`

`"type": "integer",`

`"minimum": 1,`

`"maximum": 9999999999,`

`"description": "Random seed for reproducibility of the generation."`

`},`

`"repetition_penalty": {`

```
        "type": "number",
        "minimum": 0,
        "maximum": 2,
        "description": "Penalty
for repeated tokens; higher values discourage
repetition."
    },
    "frequency_penalty": {
        "type": "number",
        "minimum": 0,
        "maximum": 2,
        "description": "Decreases
the likelihood of the model repeating the same
lines verbatim."
    },
    "presence_penalty": {
        "type": "number",
        "minimum": 0,
        "maximum": 2,
        "description": "Increases
the likelihood of the model introducing new
topics."
    }
},
"required": [
```

```
        "messages"
      ]
    },
    {
      "title":
"Ai_Cf_Meta_Llama_4_Async_Batch",
      "type": "object",
      "properties": {
        "requests": {
          "type": "array",
          "items": {
            "type": "object",
            "oneOf": [
              {
                "title":
"Ai_Cf_Meta_Llama_4_Prompt_Inner",
                "properties":
{
                  "prompt":
{
                    "type": "string",

                    "minLength": 1,
```



```
"description": "The input text prompt for the  
model to generate a response."
```

```
},
```

```
"guided_json": {
```

```
"type": "object",
```

```
"description": "JSON schema that should be  
fulfilled for the response."
```

```
},
```

```
"response_format": {
```

```
"title": "JSON Mode",
```

```
"type": "object",
```

```
"properties": {
```

```
"type": {
```

```
"type": "string",
```

```
"enum": [  
  
  "json_object",  
  
  "json_schema"  
  
],  
  
  },  
  
  "json_schema": {}  
  
    },  
    },  
    "raw": {  
  
  "type": "boolean",  
  
  "default": false,  
  
  "description": "If true, a chat template is  
not applied and you must adhere to the  
specific model's expected formatting."  
    },  
    "stream":  
  
  {
```

```
"type": "boolean",

"default": false,

"description": "If true, the response will be
streamed back incrementally using SSE, Server
Sent Events."

    },

    "max_tokens": {

        "type": "integer",

        "default": 256,

        "description": "The maximum number of tokens
to generate in the response."

    },

    "temperature": {

        "type": "number",

        "default": 0.15,
```

```
"minimum": 0,  
  
"maximum": 5,  
  
"description": "Controls the randomness of the  
output; higher values produce more random  
results."  
  
    },  
    "top_p": {  
  
"type": "number",  
  
"minimum": 0,  
  
"maximum": 2,  
  
"description": "Adjusts the creativity of the  
AI's responses by controlling how many  
possible words it considers. Lower values make  
outputs more predictable; higher values allow  
for more varied and creative responses."  
  
    },  
    "top_k": {
```

```
"type": "integer",

"minimum": 1,

"maximum": 50,

"description": "Limits the AI to choose from
the top 'k' most probable words. Lower values
make responses more focused; higher values
introduce more variety and potential
surprises."

    },
    "seed": {

"type": "integer",

"minimum": 1,

"maximum": 9999999999,

"description": "Random seed for
reproducibility of the generation."

    },
```

```
"repetition_penalty": {  
  
  "type": "number",  
  
  "minimum": 0,  
  
  "maximum": 2,  
  
  "description": "Penalty for repeated tokens;  
higher values discourage repetition."  
},  
  
"frequency_penalty": {  
  
  "type": "number",  
  
  "minimum": 0,  
  
  "maximum": 2,  
  
  "description": "Decreases the likelihood of  
the model repeating the same lines verbatim."  
},
```

```
"presence_penalty": {  
  
  "type": "number",  
  
  "minimum": 0,  
  
  "maximum": 2,  
  
  "description": "Increases the likelihood of  
the model introducing new topics."  
},  
},  
"required": [  
  "prompt"  
],  
},  
{  
  "title":  
    "Ai_Cf_Meta_Llama_4_Messages_Inner",  
  "properties":  
{  
  
  "messages": {
```

```
"type": "array",

"description": "An array of message objects
representing the conversation history.",

"items": {

  "type": "object",

  "properties": {

    "role": {

      "type": "string",

      "description": "The role of the message sender
(e.g., 'user', 'assistant', 'system',
'tool').",

    },

    "tool_call_id": {

      "type": "string",
```



```
"description": "The tool call id. If you don't  
know what to put here you can fall back to  
000000001",
```

```
"pattern": "[a-zA-Z0-9]{9}"
```

```
},
```

```
"content": {
```

```
"oneOf": [
```

```
{
```

```
"type": "string",
```

```
"description": "The content of the message as  
a string."
```

```
},
```

```
{
```

```
"type": "array",
```

```
"items": {  
  
  "type": "object",  
  
  "properties": {  
  
    "type": {  
  
      "type": "string",  
  
      "description": "Type of the content provided"  
    },  
  
    "text": {  
  
      "type": "string"  
    },  
  
    "image_url": {  
  
      "type": "object",
```

```
"properties": {  
  
  "url": {  
  
    "type": "string",  
  
    "pattern": "^data:*",  
  
    "description": "image uri with data (e.g.  
...). HTTP URL will  
not be accepted"  
  
  }  
  
}  
  
}  
  
}  
  
}  
  
},
```

```
{  
  
  "type": "object",  
  
  "properties": {  
  
    "type": {  
  
      "type": "string",  
  
      "description": "Type of the content provided"  
  
    },  
  
    "text": {  
  
      "type": "string"  
  
    },  
  
    "image_url": {  
  
      "type": "object",
```

```
"properties": {  
  
  "url": {  
  
    "type": "string",  
  
    "pattern": "^data:*",  
  
    "description": "image uri with data (e.g.  
...). HTTP URL will  
not be accepted"  
  
  }  
  
}  
  
}  
  
}  
  
}  
  
]  
]
```

```
}  
  
}  
  
},  
}
```

```
"functions": {  
  
  "type": "array",  
  
  "items": {  
  
    "type": "object",  
  
    "properties": {  
  
      "name": {  
  
        "type": "string"  
  
      },  
  
      "code": {  
  
        "type": "string"
```

```
}  
  
    },  
  
    "required": [  
  
        "name",  
  
        "code"  
  
    ],  
    },  
    },  
    "tools": {  
  
        "type": "array",  
  
        "description": "A list of tools available for  
the assistant to use.",  
  
        "items": {  
  
            "type": "object",  
  
            "oneOf": [  

```

```
{  
  
  "properties": {  
  
    "name": {  
  
      "type": "string",  
  
      "description": "The name of the tool. More  
descriptive the better."  
  
    },  
  
    "description": {  
  
      "type": "string",  
  
      "description": "A brief description of what  
the tool does."  
  
    },  
  
    "parameters": {
```



```
"type": "object",

"description": "Schema defining the parameters
accepted by the tool.",

"properties": {

  "type": {

    "type": "string",

    "description": "The type of the parameters
object (usually 'object')."

  },

  "required": {

    "type": "array",

    "description": "List of required parameter
names.",

    "items": {
```

```
"type": "string"
```

```
}
```

```
},
```

```
"properties": {
```

```
"type": "object",
```

```
"description": "Definitions of each  
parameter.",
```

```
"additionalProperties": {
```

```
"type": "object",
```

```
"properties": {
```

```
"type": {
```

```
"type": "string",
```

```
"description": "The data type of the
```

```
parameter."  
  
},  
  
"description": {  
  
  "type": "string",  
  
  "description": "A description of the expected  
parameter."  
  
}  
  
},  
  
"required": [  
  
  "type",  
  
  "description"  
  
]  
  
}
```

```
}  
  
},  
  
"required": [  
  
"type",  
  
"properties"  
  
]  
  
}  
  
},  
  
"required": [  
  
"name",  
  
"description",  
  
"parameters"
```

```
]

},

{

  "properties": {

    "type": {

      "type": "string",

      "description": "Specifies the type of tool  
(e.g., 'function')."

    },

    "function": {

      "type": "object",

      "description": "Details of the function  
tool."

    },
```

```
"properties": {  
  
  "name": {  
  
    "type": "string",  
  
    "description": "The name of the function."  
  
  },  
  
  "description": {  
  
    "type": "string",  
  
    "description": "A brief description of what  
the function does."  
  
  },  
  
  "parameters": {  
  
    "type": "object",  
  
    "description": "Schema defining the parameters
```

```
accepted by the function.",

"properties": {

"type": {

"type": "string",

"description": "The type of the parameters
object (usually 'object')."

},

"required": {

"type": "array",

"description": "List of required parameter
names.",

"items": {

"type": "string"

}

}
```

```
},  
  
"properties": {  
  
  "type": "object",  
  
  "description": "Definitions of each  
parameter.",  
  
  "additionalProperties": {  
  
    "type": "object",  
  
    "properties": {  
  
      "type": {  
  
        "type": "string",  
  
        "description": "The data type of the  
parameter."  
  
      },  

```



```
"description": {  
  
  "type": "string",  
  
  "description": "A description of the expected  
parameter."  
  
}  
  
},  
  
"required": [  
  
  "type",  
  
  "description"  
  
]  
  
}  
  
}
```

```
"required": [
```

```
"type",
```

```
"properties"
```

```
]
```

```
}
```

```
},
```

```
"required": [
```

```
"name",
```

```
"description",
```

```
"parameters"
```

```
]
```

```
}
```

```

},
"required": [
"type",
"function"
]
}
]
}
},
"response_format": {
"title": "JSON Mode",
"type": "object",
"properties": {
"type": {

```

```
"type": "string",

"enum": [

"json_object",

"json_schema"

]

},

"json_schema": {}

},

"guided_json": {

"type": "object",

"description": "JSON schema that should be
fulfilled for the response."

},

"raw": {
```

```
"type": "boolean",

"default": false,

"description": "If true, a chat template is
not applied and you must adhere to the
specific model's expected formatting."
    },
    "stream":
{

"type": "boolean",

"default": false,

"description": "If true, the response will be
streamed back incrementally using SSE, Server
Sent Events."
    },

    "max_tokens": {

"type": "integer",
```

```
"default": 256,  
  
"description": "The maximum number of tokens  
to generate in the response."  
  
    },  
  
"temperature": {  
  
"type": "number",  
  
"default": 0.15,  
  
"minimum": 0,  
  
"maximum": 5,  
  
"description": "Controls the randomness of the  
output; higher values produce more random  
results."  
  
    },  
    "top_p": {  
  
"type": "number",
```

```
"minimum": 0,  
  
"maximum": 2,  
  
"description": "Adjusts the creativity of the  
AI's responses by controlling how many  
possible words it considers. Lower values make  
outputs more predictable; higher values allow  
for more varied and creative responses."  
    },  
    "top_k": {  
  
"type": "integer",  
  
"minimum": 1,  
  
"maximum": 50,  
  
"description": "Limits the AI to choose from  
the top 'k' most probable words. Lower values  
make responses more focused; higher values  
introduce more variety and potential  
surprises."  
    },
```

```
"seed": {  
  
  "type": "integer",  
  
  "minimum": 1,  
  
  "maximum": 9999999999,  
  
  "description": "Random seed for  
reproducibility of the generation."  
},  
  
"repetition_penalty": {  
  
  "type": "number",  
  
  "minimum": 0,  
  
  "maximum": 2,  
  
  "description": "Penalty for repeated tokens;  
higher values discourage repetition."  
},  
  
"frequency_penalty": {
```



```
"type": "number",

"minimum": 0,

"maximum": 2,

"description": "Decreases the likelihood of
the model repeating the same lines verbatim."
    },

"presence_penalty": {

"type": "number",

"minimum": 0,

"maximum": 2,

"description": "Increases the likelihood of
the model introducing new topics."
    }
},
"required": [
    "messages"
```

```
}  
  ]  
}  
  }  
},  
  "required": [  
    "requests"  
  ]  
}
```

Was this helpful?



Previous

← Dashboard

Next

Workers Bindings →

Resources

Support

[API](#)[Help Center](#)[New to Cloudflare?](#)[System Status](#)[Directory](#)[Compliance](#)[Sponsorships](#)[GDPR](#)[Open Source](#)

Company

Tools

[cloudflare.com](#)[Cloudflare Radar](#)[Our team](#)[Speed Test](#)[Careers](#)[Is BGP Safe Yet?](#)[RPKI Toolkit](#)[Certificate Transparency](#)

Community

[!\[\]\(1ed10657a19f9137278430c48fd18626_img.jpg\) X](#)[!\[\]\(4c9516d2c24d0d513bc9f84c2e013d65_img.jpg\) Discord](#)[!\[\]\(a25a22d88c5882f4a20f36103df86562_img.jpg\) YouTube](#)[!\[\]\(06b7456efb47d301bca6298603e7f4fc_img.jpg\) GitHub](#)

- [Report Security Issues](#)
- [Trademark](#)
-  [Cookie Preferences](#)